GameScope: Predicting Video Game Success from Market and Review Data

Motivation

The goal of this project is to investigate which factors are associated with the success of video games and whether game success can be predicted using structured market and review-related features. Video games differ in price, genre, release timing, user engagement, and player reception, and these differences may help explain why some games perform better than others. This topic is especially interesting because video game success is influenced by both measurable and less visible factors, which makes it suitable for a data science project involving exploration, statistical testing, and predictive modeling.

Data Source

The main data used in this project comes from publicly available Steam-related datasets. The core dataset, `steam.csv`, contains game-level information such as name, release date, genres, Steam tags, positive ratings, negative ratings, average playtime, owners, price, achievements, required age, and language-related variables. To enrich the project, additional Steam-related tag information was also considered, and several engineered variables were created from the original data.

The data was collected from publicly available CSV files and loaded into Python using pandas. After collection, the data was cleaned by removing duplicates, handling missing values, converting dates, and selecting relevant columns for analysis.

Data Preparation and Feature Engineering

Several derived variables were created in order to support both analysis and machine learning. These included:

- `total_reviews`
- `positive_ratio`
- `release_year`
- `game_age`
- `is_free`
- `genre_count`
- `tag_count`
- `has_achievements`
- `log_playtime`
- `log_total_reviews`

These features were created to better capture game characteristics beyond the original raw columns.

Exploratory Data Analysis

Exploratory data analysis was used to understand the structure of the dataset and identify broad patterns. Distribution plots showed that most games are concentrated in the lower price range, while only a smaller number of games are sold at higher prices. The positive review ratio distribution was skewed toward higher values, suggesting that many games receive mostly positive feedback. Review counts were highly uneven, with a few games receiving very large numbers of reviews.

Scatterplots and grouped analyses suggested that simple linear relationships between price and positive review ratio were weak. Average playtime and release year also showed some variation, but no single variable appeared to strongly explain game success on its own. These results suggested that the relationship between available features and user satisfaction is likely complex.

Hypothesis Testing

Three hypothesis-testing analyses were performed. First, a t-test was used to compare the mean positive review ratio of free-to-play and paid games. Second, Pearson correlation was used to test the relationship between price and positive review ratio. Third, another correlation analysis examined the relationship between total review count and positive review ratio.

These tests provided an initial statistical view of the data. They showed whether some relationships were statistically significant, but they also suggested that significance does not necessarily imply strong predictive power.

Machine Learning Methods

To predict game success, measured by `positive_ratio`, two regression models were applied:

- Linear Regression
- Random Forest Regressor

A baseline feature set was first used, including price, average playtime, release year, game age, and free-to-play status. After receiving feedback, an improved model was developed using additional engineered features such as required age, English language availability, achievements, genre count, tag count, log-transformed playtime, and log-transformed total reviews.

The models were evaluated using MAE, RMSE, and R^2.

Findings

The machine learning results showed that predictive performance was limited. Although the improved model performed somewhat better than the baseline model, the R^2 values remained relatively weak. This indicates that the available structured features explain only a limited portion of the variation in positive review ratio.

This finding is meaningful. It suggests that video game success, especially user satisfaction, depends on many factors that are not fully captured by the current dataset. The feature importance results showed that some variables, such as review-related scale measures and price-related characteristics, contributed more than others, but overall prediction remained difficult.

Why the R^2 Scores Were Weak

The relatively weak R^2 scores can be explained by the nature of the problem. Positive review ratio is affected not only by measurable metadata such as price or playtime, but also by factors such as game quality, player expectations, genre-specific audience behavior, developer reputation, community reception, timing, market competition, and many other hidden influences.

Even after adding more engineered variables, the models still had limited explanatory power. This suggests that structured numerical data alone is not enough to fully explain success in video games.

Limitations and Future Work

This project has several limitations. First, the available data is mostly structured and does not include richer textual or behavioral information. Second, some important variables, such as detailed review text, developer history, marketing strength, update frequency, and community activity, are not present. Third, success itself is multidimensional, and using positive review ratio as the target captures only one aspect of it.

Future work could improve the project in several ways. Text analysis on game descriptions or review text could be added. More metadata about developers, publishers, or player communities could also be included. Alternative targets such as popularity classes or owner ranges could be tested. In addition, more advanced models and model tuning could be explored.

Conclusion

This project applied the data science pipeline to the question of video game success prediction using Steam-related data. The project included data collection, cleaning, exploratory data analysis, hypothesis testing, feature engineering, and machine learning. While the predictive results were limited, the project still provided useful insight: game success is a complex phenomenon that cannot be fully explained by a small set of structured variables. This makes the weak R^2 scores not just a limitation, but also an important finding of the analysis.